

Project Report: AI-Driven Skill Gap Analysis (2026 Market Trend)

Project Title: 2026 CS Engineer Skill Gap & Market Trend
CLI Analyzer

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Branch : CSE(AIML)

Course Code : CSA2001 – Introduction to AIML

Slot: A21+A22+A23

Tech Stack: Python, Scikit-Learn, NumPy, Pandas, Colorama

1. Executive Summary

In 2026, the traditional software engineering role has undergone a paradigm shift. With the rise of **Agentic AI** and **Model Context Protocol (MCP)**, the demand for "pure coders" has decreased, while the demand for "AI Architects" has surged. This project provides a lightweight, terminal-based solution to help engineers quantify their alignment with these 2026 market shifts using mathematical vector similarity models.

2. Problem Statement

Many engineers rely on outdated curriculum standards that do not account for the **53% talent deficit** in Agentic AI and the **55% deficit** in Cloud-Native Engineering.

- **The Challenge:** Engineers often don't know *exactly* how far behind they are.
 - **The Solution:** A tool that compares a user's skill profile against a dynamic "Ideal Market Vector" and provides a numerical "Market Alignment Score."
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3. Market Research & Statistical Basis (2026 Trends)

The project utilizes integrated data points reflecting the 2026 landscape:

Specialization	YoY Growth	Criticality	Salary Premium (Avg)
Agentic AI	+156%	High	₹25L - ₹45L
Cybersecurity	+35%	Medium-High	₹18L - ₹30L
Cloud Native	+28%	High	₹16L - ₹28L

4. Technical Architecture

4.1 Vector Space Modeling (VSM)

The core logic treats every job role and every candidate as a vector in a 5-dimensional space ($\mathbb{R}^5$).

The dimensions are:

1. **Agentic AI**
2. **MLOps**
3. **Cloud Native**
4. **Cybersecurity**
5. **DSA**

4.2 The Mathematical Model

The project implements **Cosine Similarity** to determine the "closeness" of the user's skills to the market's needs. Unlike Euclidean distance, Cosine Similarity focuses on the **pattern and balance** of skills rather than just the raw magnitude.

$$\text{Readiness Score} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

4.3 Data Structures

- **Pandas Data Frames:** Used to manage and display market statistics.
 - **NumPy Arrays:** Used for high-speed vector calculations.
 - **Dictionary Mapping:** Used for storing role-specific benchmarks (Ideal Vectors).
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5. Implementation Details

5.1 CLI Interface

The program uses a **Command Line Interface (CLI)** for maximum efficiency. It employs Colorama to provide visual feedback:

- **Green:** High alignment (>70%).
- **Red:** Critical gaps identified (>20% deficit in a specific skill).

5.2 User Workflow

1. **Initialization:** The user views the 2026 market summary.
 2. **Target Selection:** The user selects a role (e.g., AI/ML Engineer).
 3. **Self-Assessment:** The user inputs their proficiency (0.0 to 1.0).
 4. **Vector Processing:** The system runs the similarity algorithm.
 5. **Report Generation:** The system outputs the Alignment Score and highlights specific "Critical Gaps."
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6. Results and Analysis

The model successfully identifies specific career trajectories. For instance, a candidate with a high DSA score but low Cloud-Native score is flagged as having a "**Legacy Profile**" and is directed toward modern infrastructure training to increase their market value.

7. Conclusion & Future Scope

This project serves as a proof-of-concept for AI-augmented career counseling.

Future Roadmap:

- **Resume Parsing:** Integrating PyMuPDF to extract skills automatically from a PDF resume.
- **Dynamic Scrapping:** Using BeautifulSoup to scrape live job postings from LinkedIn/Indeed to update the "Market Vectors" in real-time.
- **API Integration:** Connecting to Coursera or Udemy APIs to suggest specific courses for identified gaps.